

## Slide-1

# TOPIC

## Design and analysis of a tunable eccentric bearing block mechanism for vibration control in spur gear systems with angular misalignment error

Journal of Measurement Volume 256, Part A, 1 December 2025,  
118126

### **Presented by-**

Bharad pushkar & Joydeep Debbarma

**Subject-**Dynamics Control System

### Authors-

Desheng Zou, Liming Wang, Bo  
Zou, Wennian Yu, Yimin Shao,  
Mechefske Chris

# Index

- 1) Introduction
- 2) Methodology
- 3) Results
- 4) Discussion
- 5) References

## Slide-3

# Introduction

- Gear transmission systems are widely used for their compactness and high transmission.
- Angular misalignment error causes uneven load and vibrations.
- Existing methods-Tooth modification, Passive isolation, active control.
- Limitation of these methods-Fixed range, no load correction, limited adaptability.
- To address the issues, we introduce a tunable eccentric bearing block(TEBB).

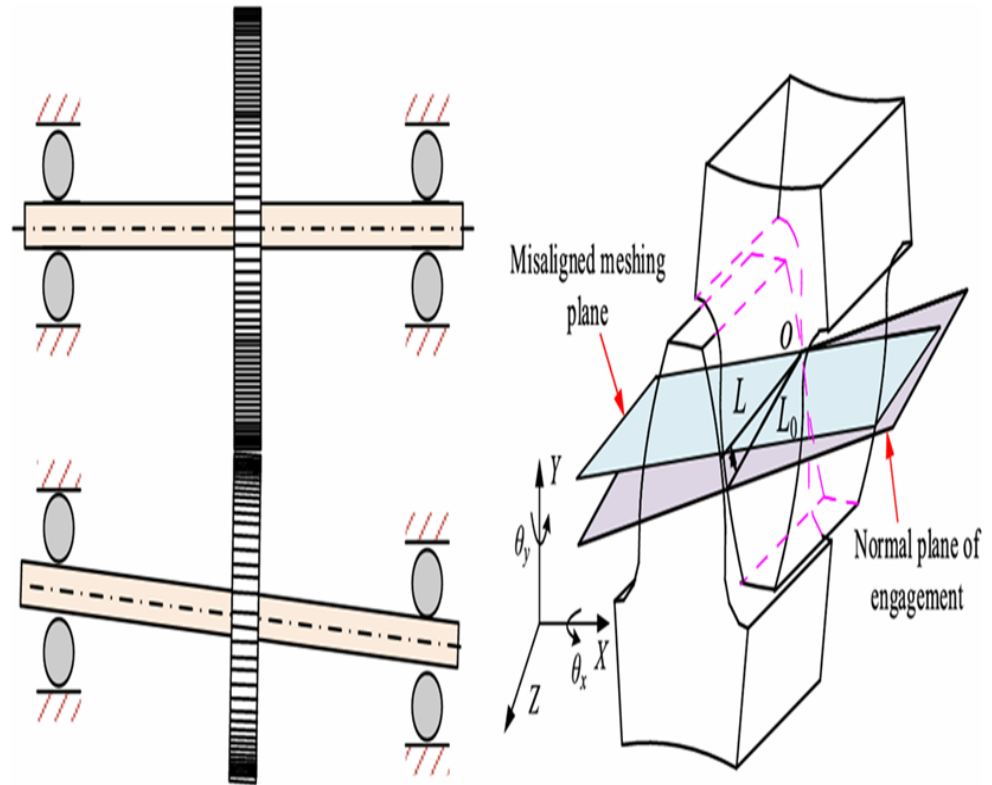


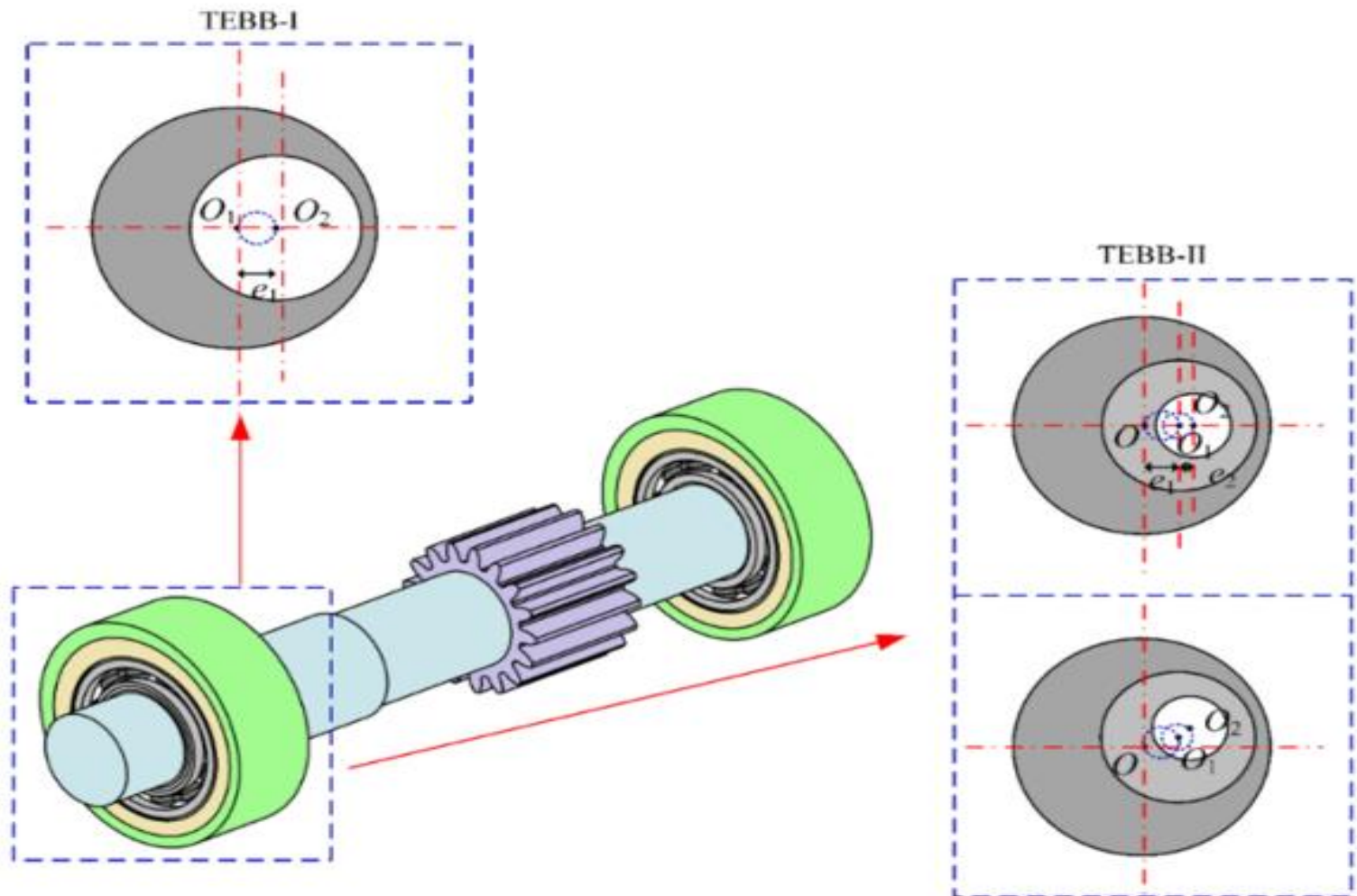
Fig-1 Diagram of shaft misalignment and gear angular misalignment error[Liming Wang]

Slide-4



Fig -2 tunable bearing [<http://tunablebearing.com>]

# Slide-5



- Fig. 2. Diagram of the mechanism of double end TEBB-I and TEBB-II.[Liming wang]

# Methodology

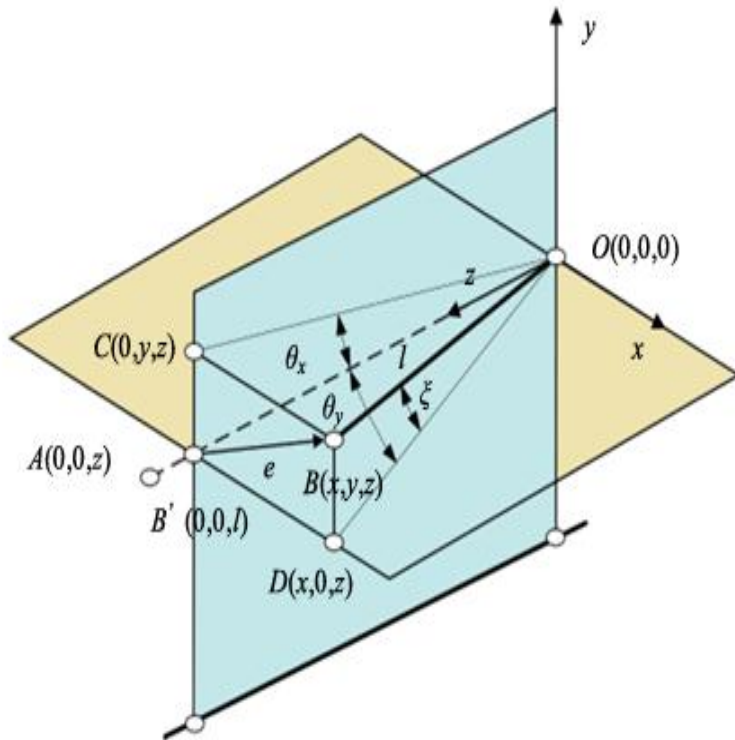


Fig. 3. The diagram of angle relationship considering angular misalignment error. [4]

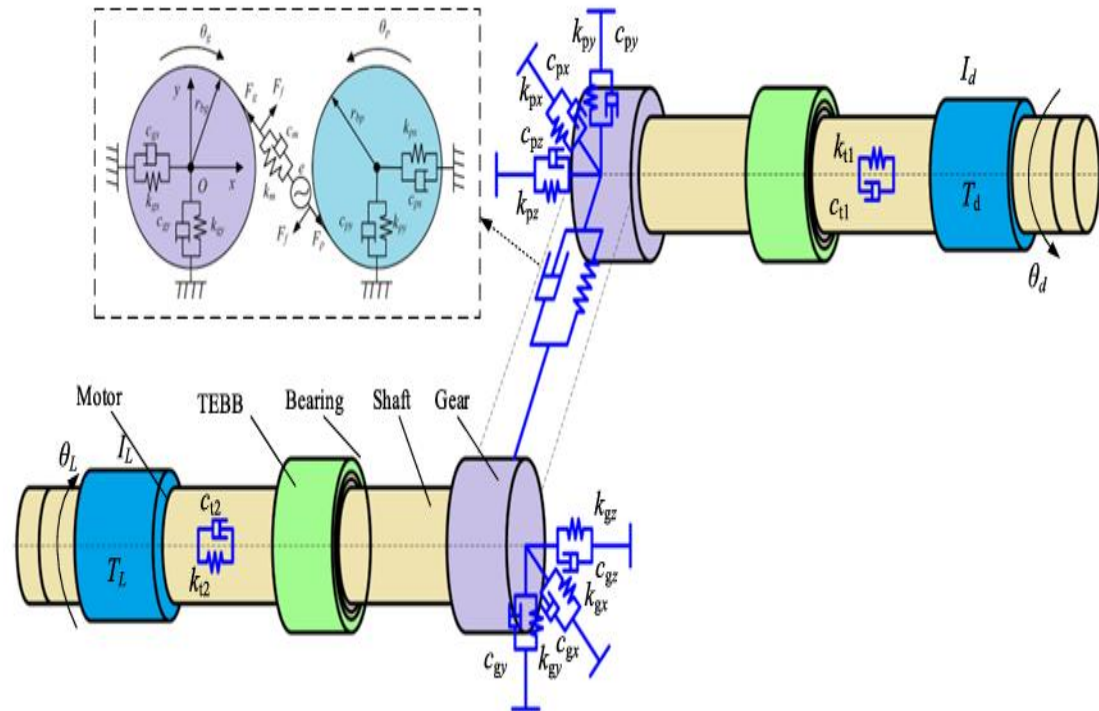


Fig. 4. 10-DOF dynamic model [17]

# Methodology

$\alpha_{1i}$  = Meshing angle of the  $i$ -th gear teeth

$b_j$  = Tooth width of each slice gear

$d_i$  = The distance from the tooth force to the origin

$e_i$  = Eccentricity of TEBB

$F_f$  = Frictional force

$F_m$  = Dynamic meshing force

$F_{mis}$  = Tooth force

$K_{bi}/K_{si}/K_{ai}/K_{hi}/K_{fi}$  = Stiffness of bending/shear/axial compression/contact/fillet foundation of gear slice

$K_{ei}$  = Equivalent stiffness on the meshing line

$l$  = Bearing hole span at both ends of gearbox shaft

$m_p/m_g$  = Equivalent mass of pinion and gear

$T_d$  = Torque of power element

$T_L$  = Torque of load element

$r_{bg}/r_{bp}$  = The pitch circle radius of gear and pinion

$W$  = Tooth width

$\theta_1/\theta_2$  = Angle between the TEBB and the position direction of  $x$  axis

$\theta_x/\theta_y$  = Angular misalignment error in  $x/y$  direction

$\delta_{mis}$  = Contact deformation

$r_{int}$  = The radius of the inner hole

$r_f$  = The radius of the base circle

$u_f$  = Effective lever at the midpoint of the tooth attachment region

$S_f, L^*, M^*, P^*, Q^*$  = The coefficients of tooth fillet-foundation stiffness

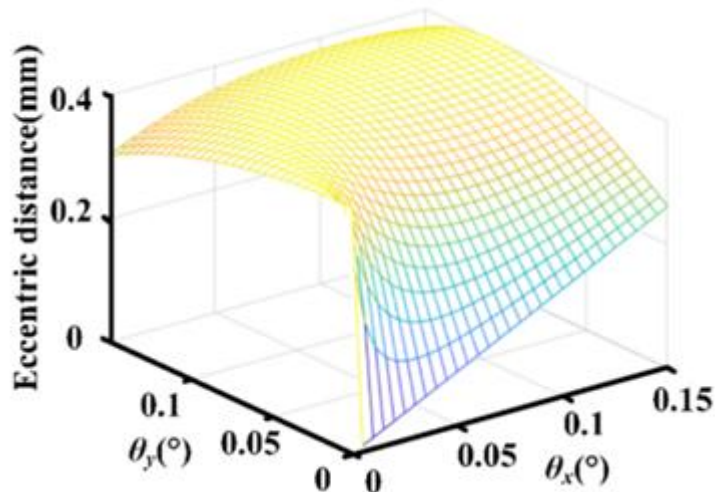


Fig. 5. Axis parallel misalignment considering angular misalignment error under TEBB-I[3]

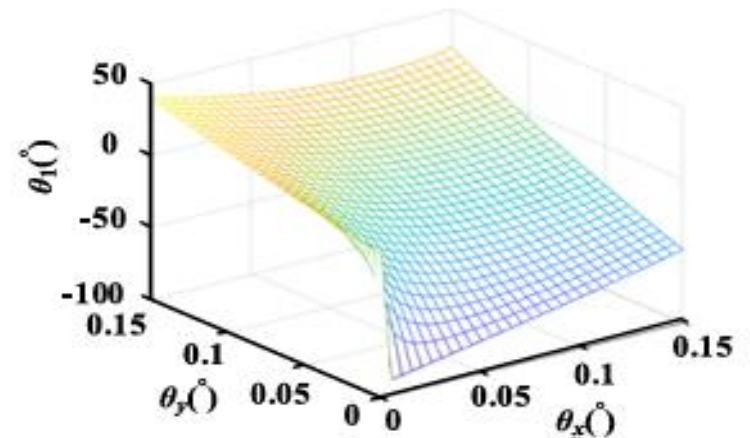


Fig. 6. The rotation angle of the right TEBB[3]

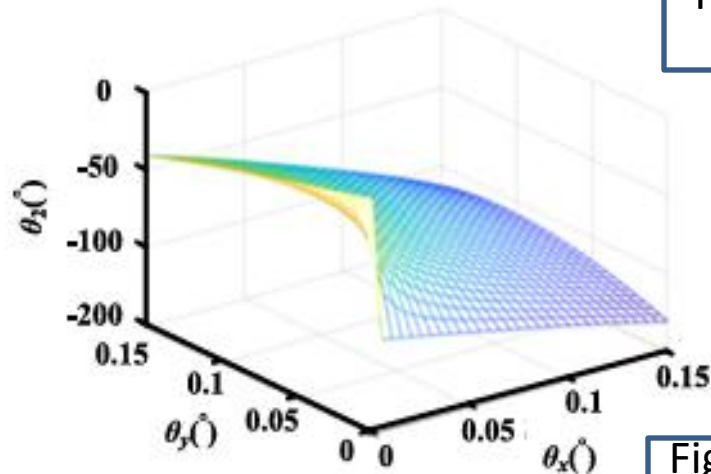


Fig. 7. The rotation angle of the left TEBB[3]

# Methodology

$$e_m = L_o \times \tan[\arctan(\tan\theta_y \sin\alpha_n + \tan\theta_x \cos\alpha_n)]$$

$$x^2 + y^2 + z^2 = l^2, \tan_y = \frac{x}{z}, \tan x = \frac{y}{z}$$

$$\frac{1}{k_{ei}} = \frac{1}{k_{bpi}} + \frac{1}{k_{spi}} + \frac{1}{k_{api}} + \frac{1}{k_{fpi}} + \frac{1}{k_{bgi}} + \frac{1}{k_{sgi}} + \frac{1}{k_{agi}} + \frac{1}{k_{fgi}}$$

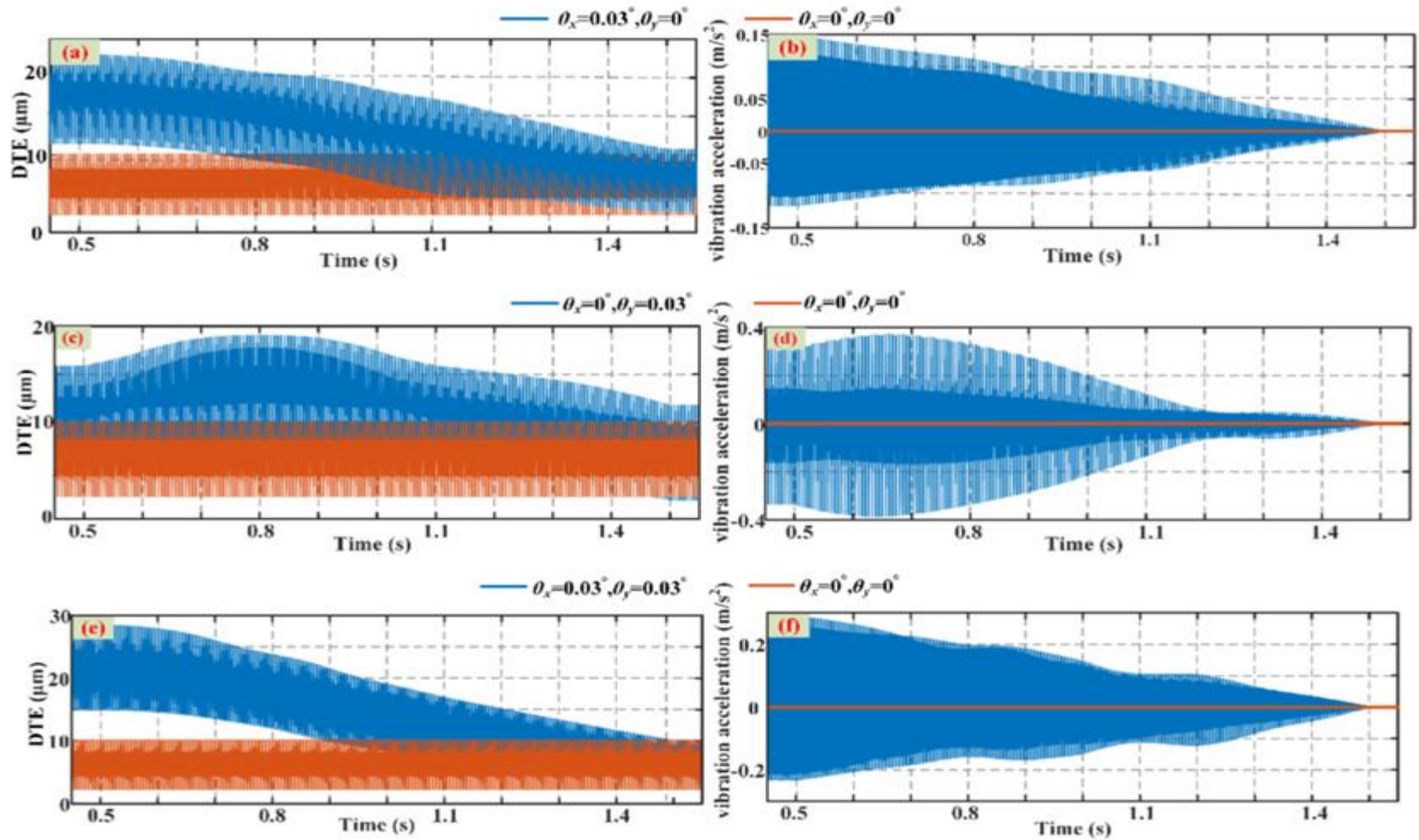
$$\begin{aligned} \delta mis &= (\Delta x \ \Delta y \ \Delta z) M_z \begin{pmatrix} \cos\alpha \\ \sin\alpha \\ 0 \end{pmatrix} + (\theta_p \ \theta_g \ 0) M_z \begin{pmatrix} r_{bp} \\ r_{bg} \\ 0 \end{pmatrix} \\ &= \cos\theta_y \cos\alpha + \Delta y (\sin\xi \sin\theta_y \cos\alpha + \cos\xi \sin\alpha) + \\ &\Delta z (\cos\xi \sin\theta_y \cos\alpha - \sin\xi \sin\alpha) + r_{bp} \theta_p \cos\theta_y + \theta_g (r_{bp} \sin\xi \sin\theta_y + r_{bg} \cos\xi) \end{aligned}$$

$$I_d \theta_d - k_{t1} (\theta_d - \theta_p) - c_{t1} (\dot{\theta}_d - \dot{\theta}_p) = T_d$$

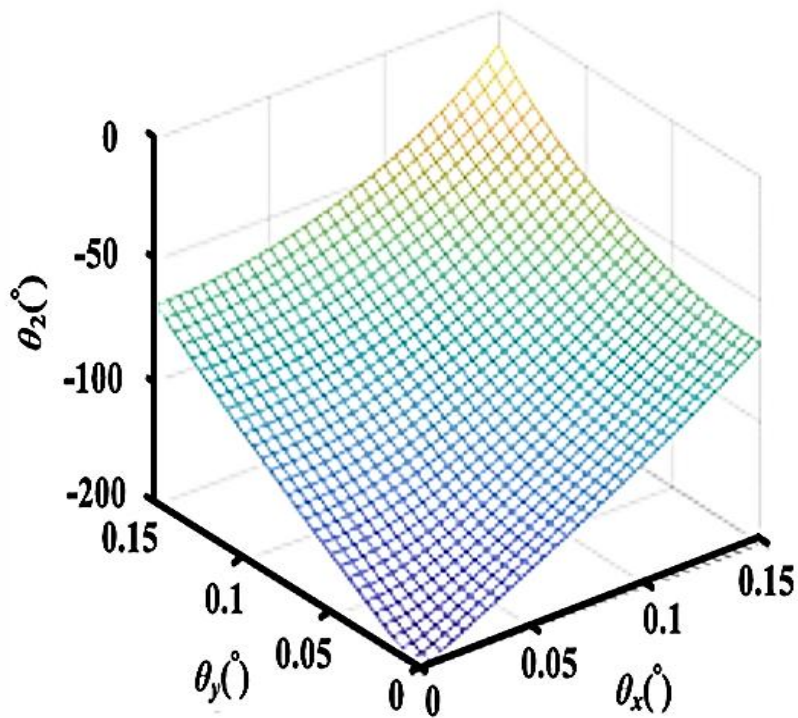
# Results

- Performance:
  - DTE reduced by ~50%
  - Restored full tooth width contact
  - Improved TVMS and contact line length
- Dynamic Behavior:
  - DTE and vibration decrease during adjustment
  - Equal-duration control gives smoother motion

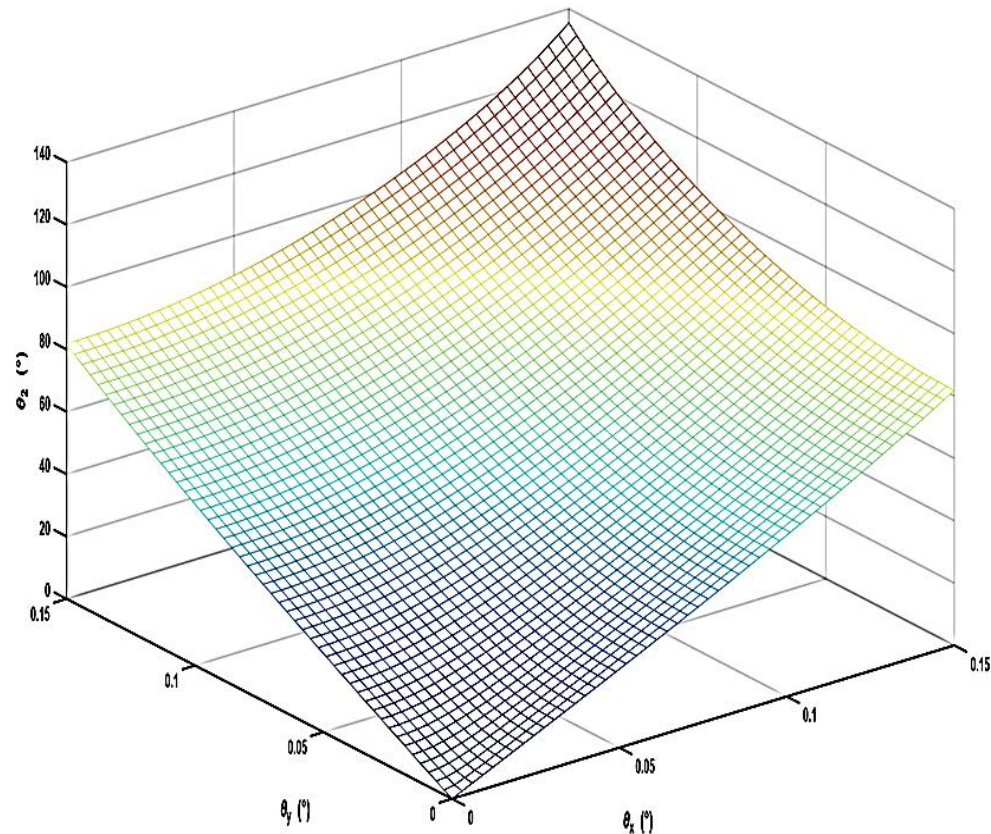
## Results



• Fig. 8. Adjustment process of angular misalignment error in different direction[Liming Wang]



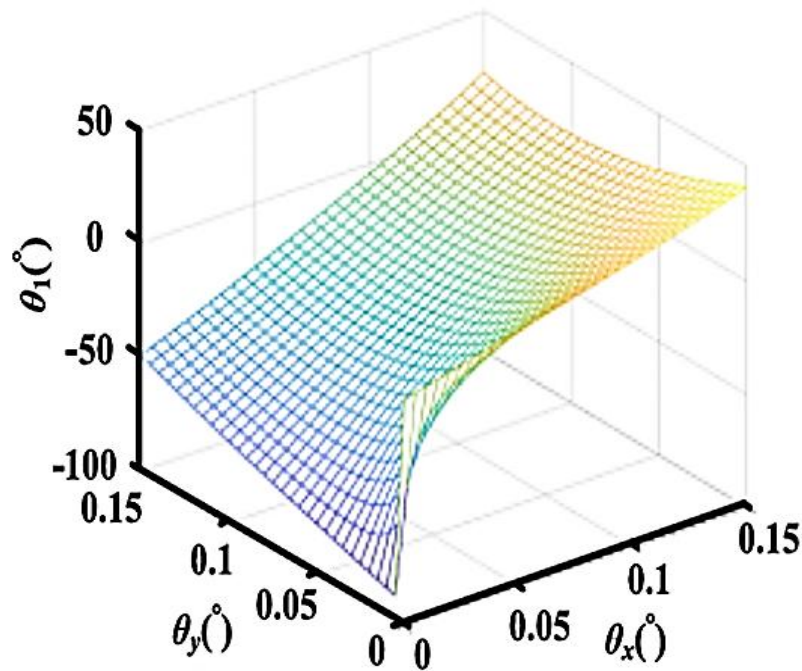
Actual Output



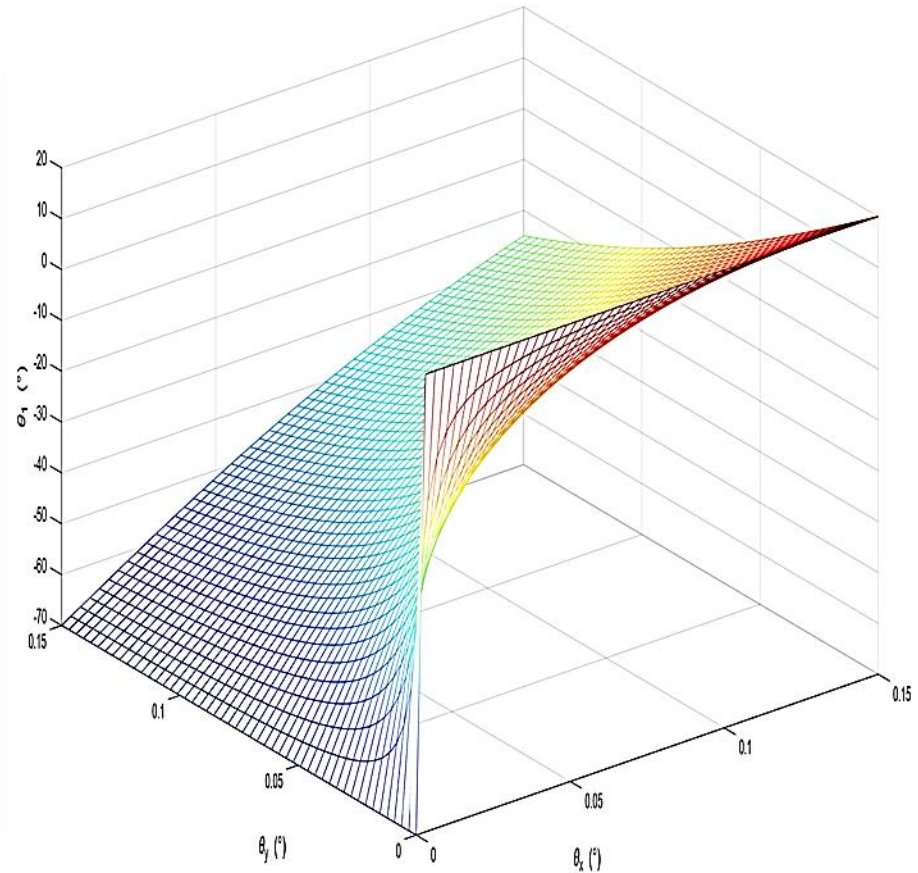
Matlab Output

Fig. 13(b) Rotation angle of inner eccentric block

Slide-13



Actual Output



Matlab Output

Fig.13(a) Rotation angle of outer eccentric block

## Slide-14

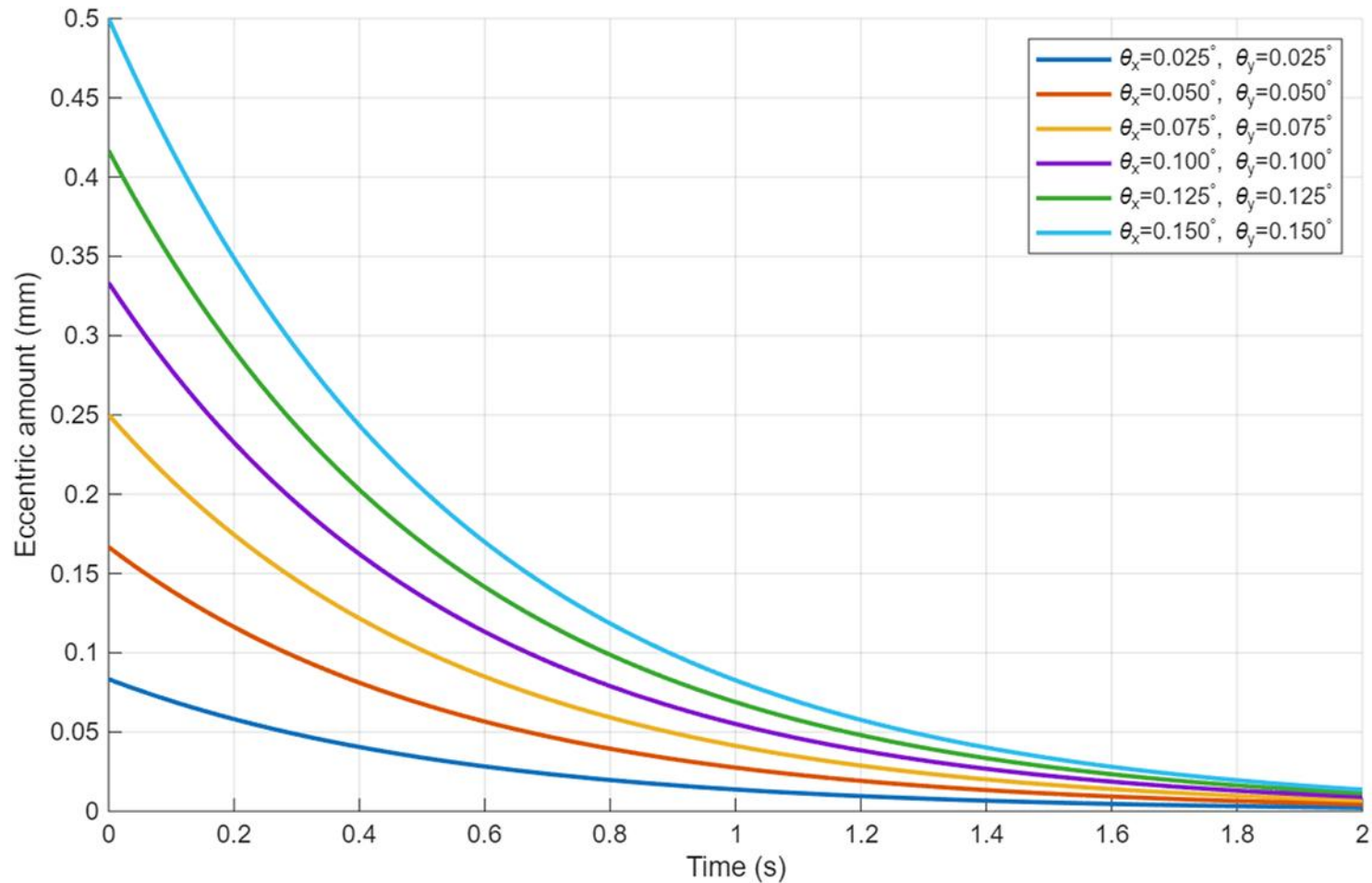


Fig. 16(a). When there is angular misalignment error in the comprehensive direction, the inner and outer eccentric blocks rotate for the same time: eccentricity change curve

## Slide-15

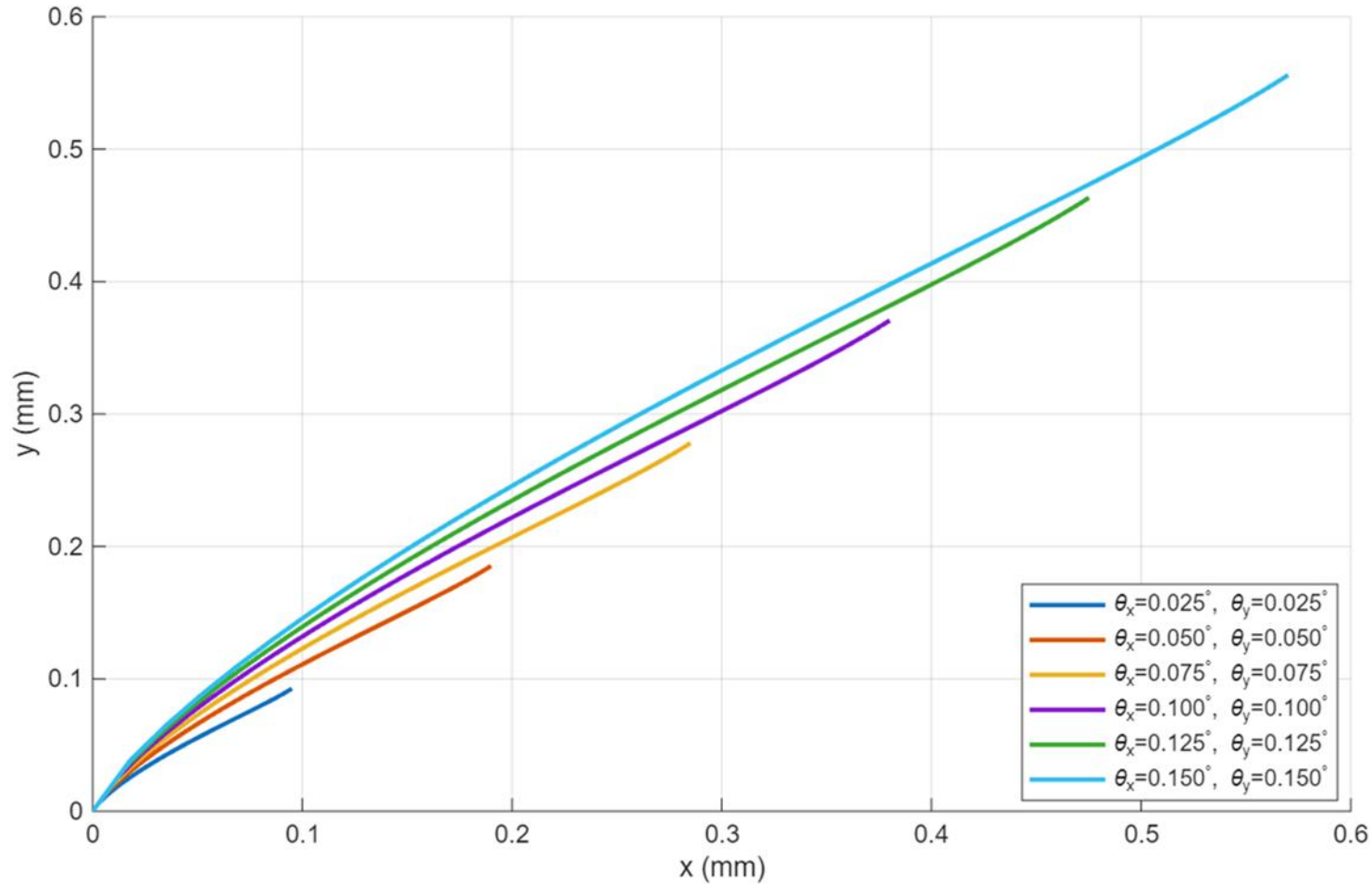


Fig. 16(b). When there is angular misalignment error in the comprehensive direction, the inner and outer eccentric blocks rotate for the same time: movement track of inner circle center of inner eccentric block

## Slide-16

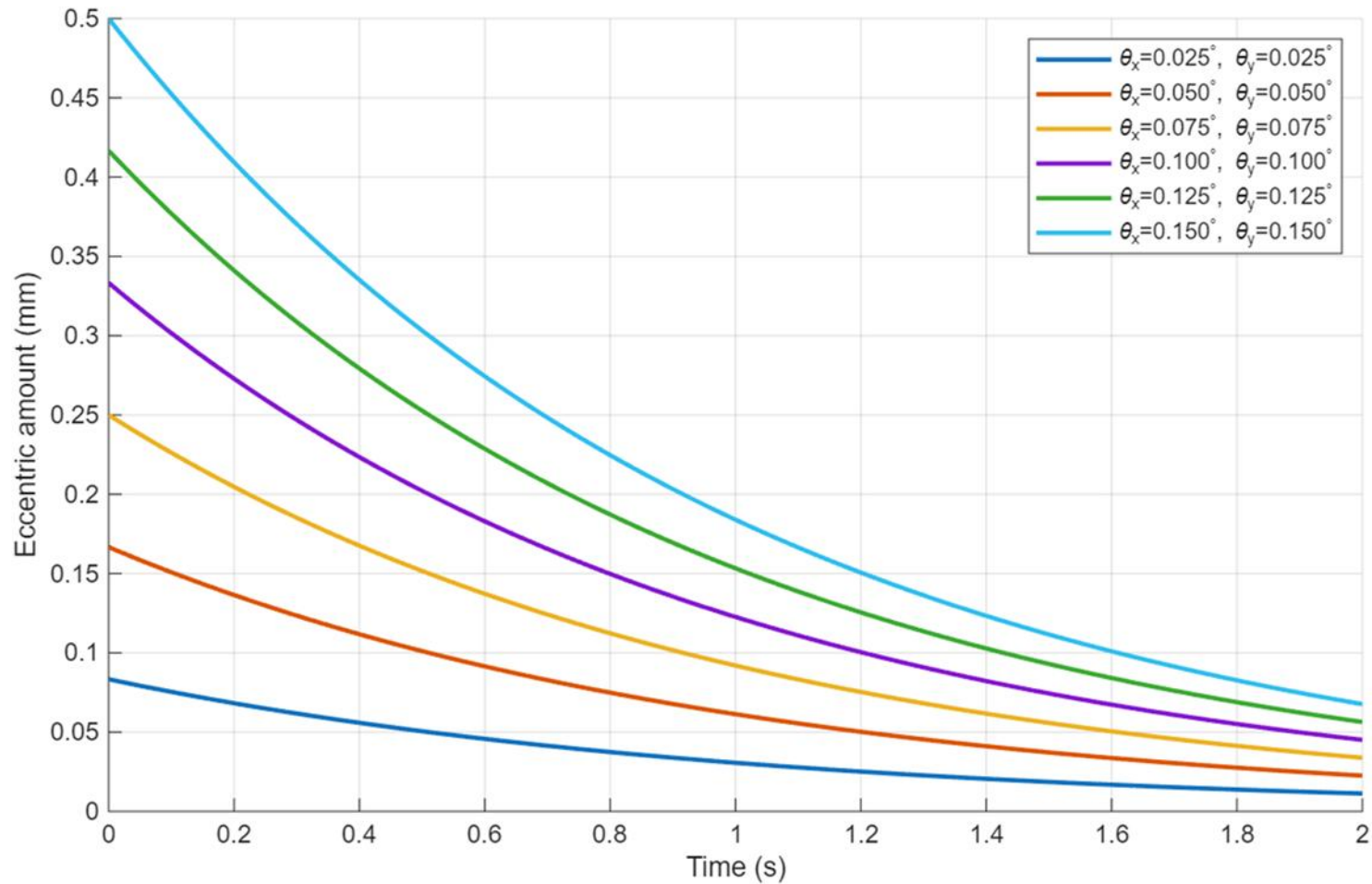


Fig. 16(c). when there is angular misalignment error in the comprehensive direction, the inner and outer eccentric blocks rotate at the same speed:  
Eccentricity change curve

## Slide-17

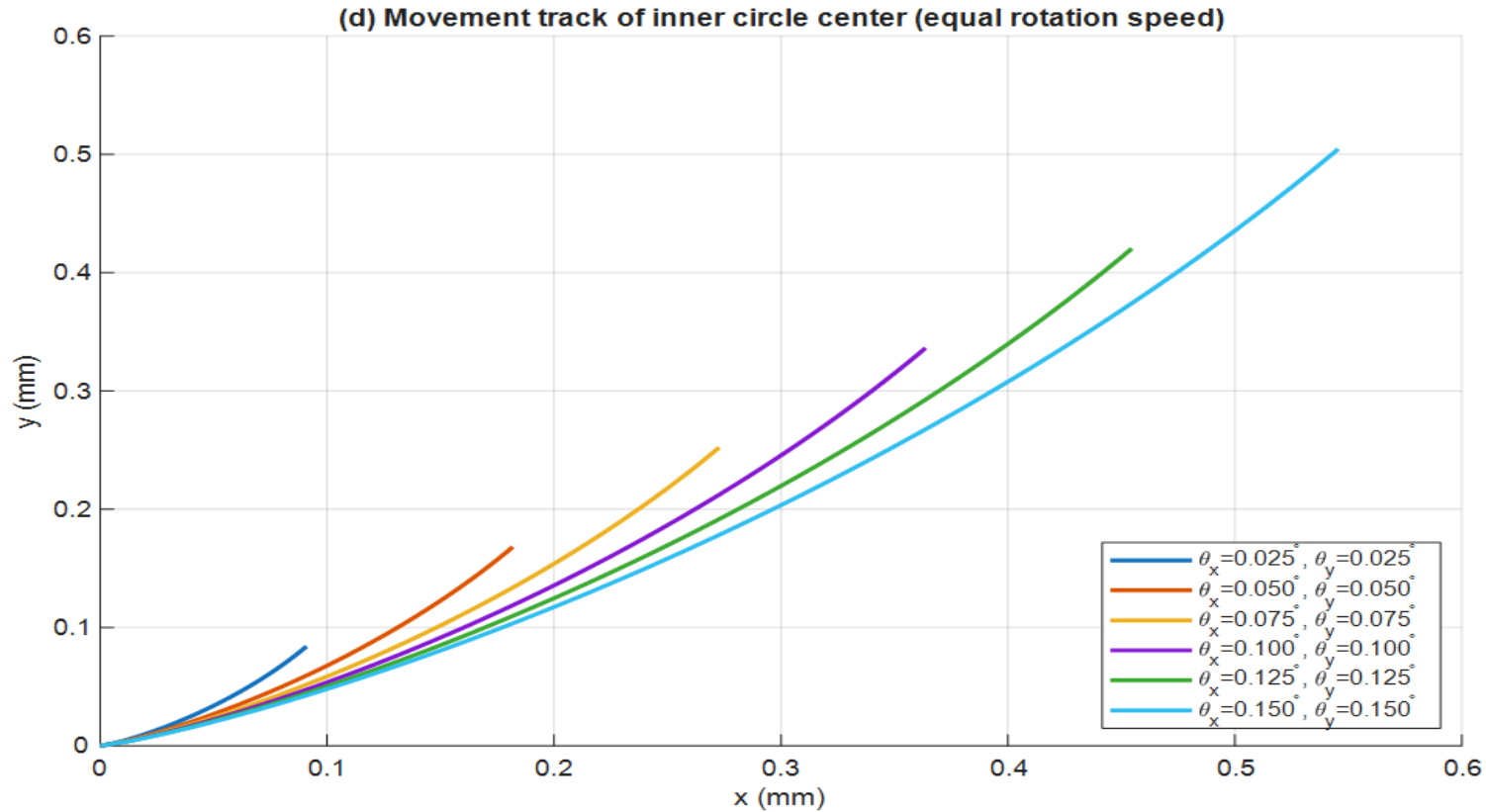


Fig. 16(d).when there is angular misalignment error in the comprehensive direction, the inner and outer eccentric blocks rotate at the same speed: Movement track of inner circle center of inner eccentric block.

# Actual Output

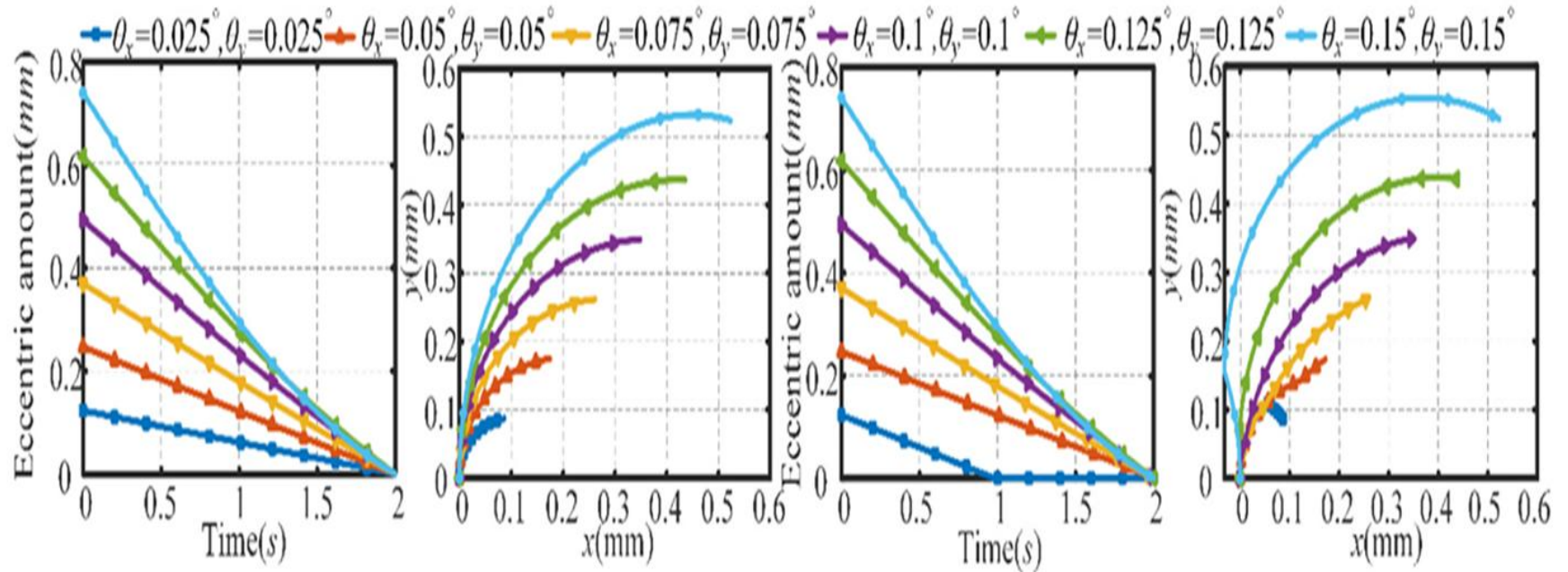
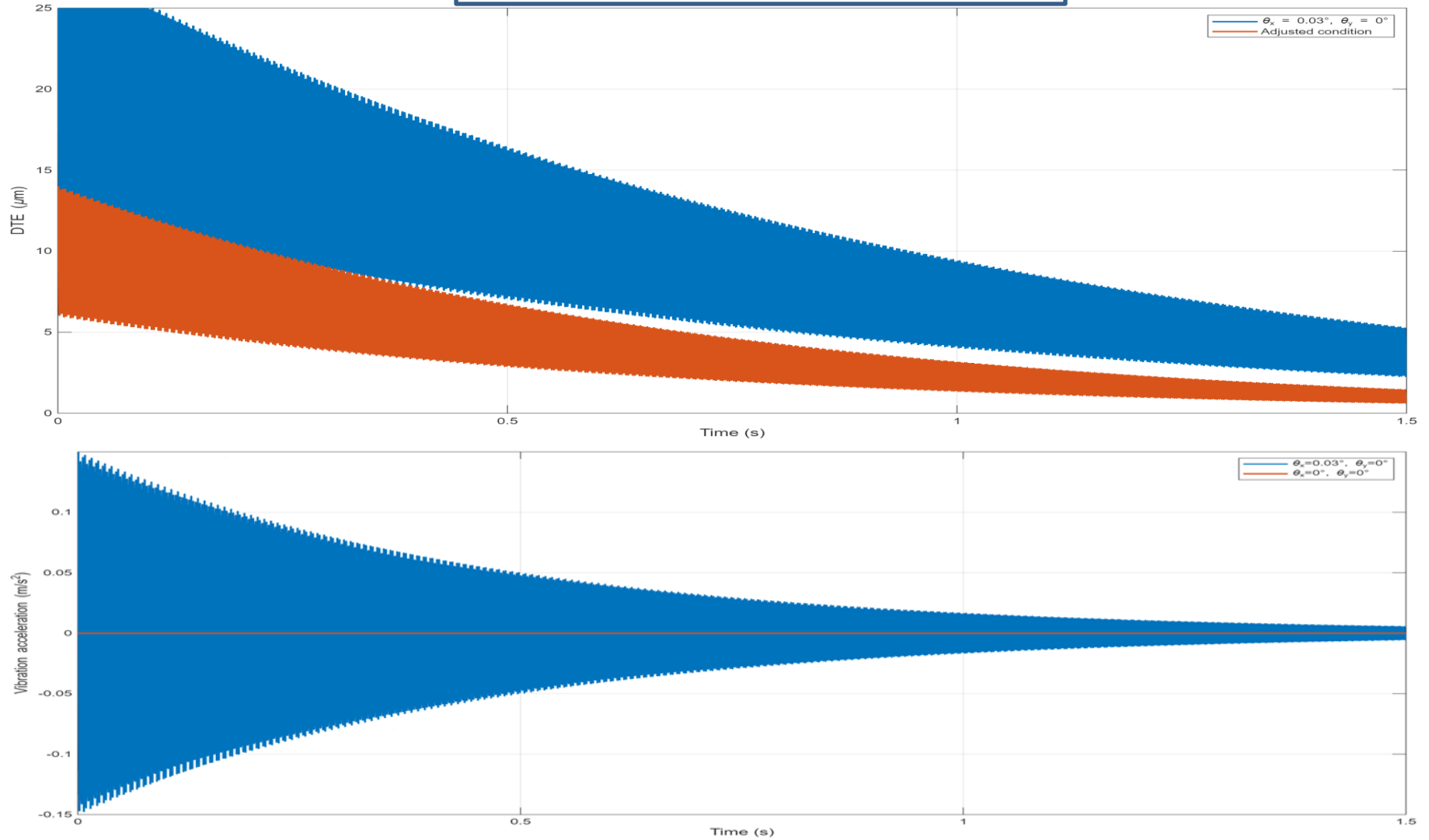


Fig. 16. When there is angular misalignment error in the comprehensive direction, the inner and outer eccentric blocks rotate for the same time: (a) eccentricity change curve, (b) movement track of inner circle center of inner eccentric block. when there is angular misalignment error in the comprehensive direction, the inner and outer eccentric blocks rotate at the same speed: (c) Eccentricity change curve, (d) Movement track of inner circle center of inner eccentric block.



- Fig21a , b-Adjustment process of angular misalignment error in x direction

## Actual Code

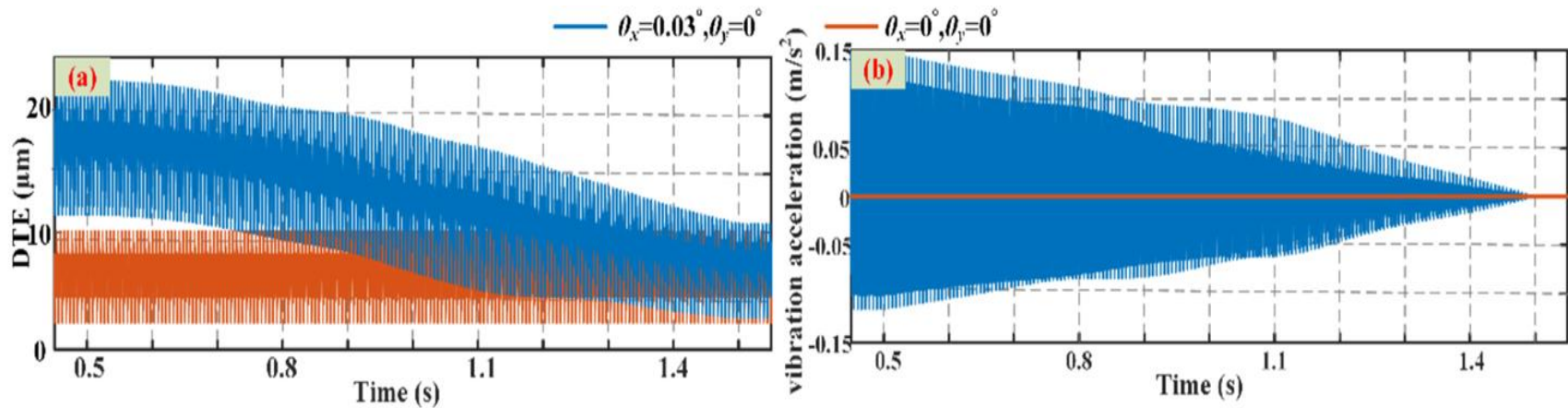


Fig. 21. Adjustment process of angular misalignment error in different direction. (a), (b) x direction

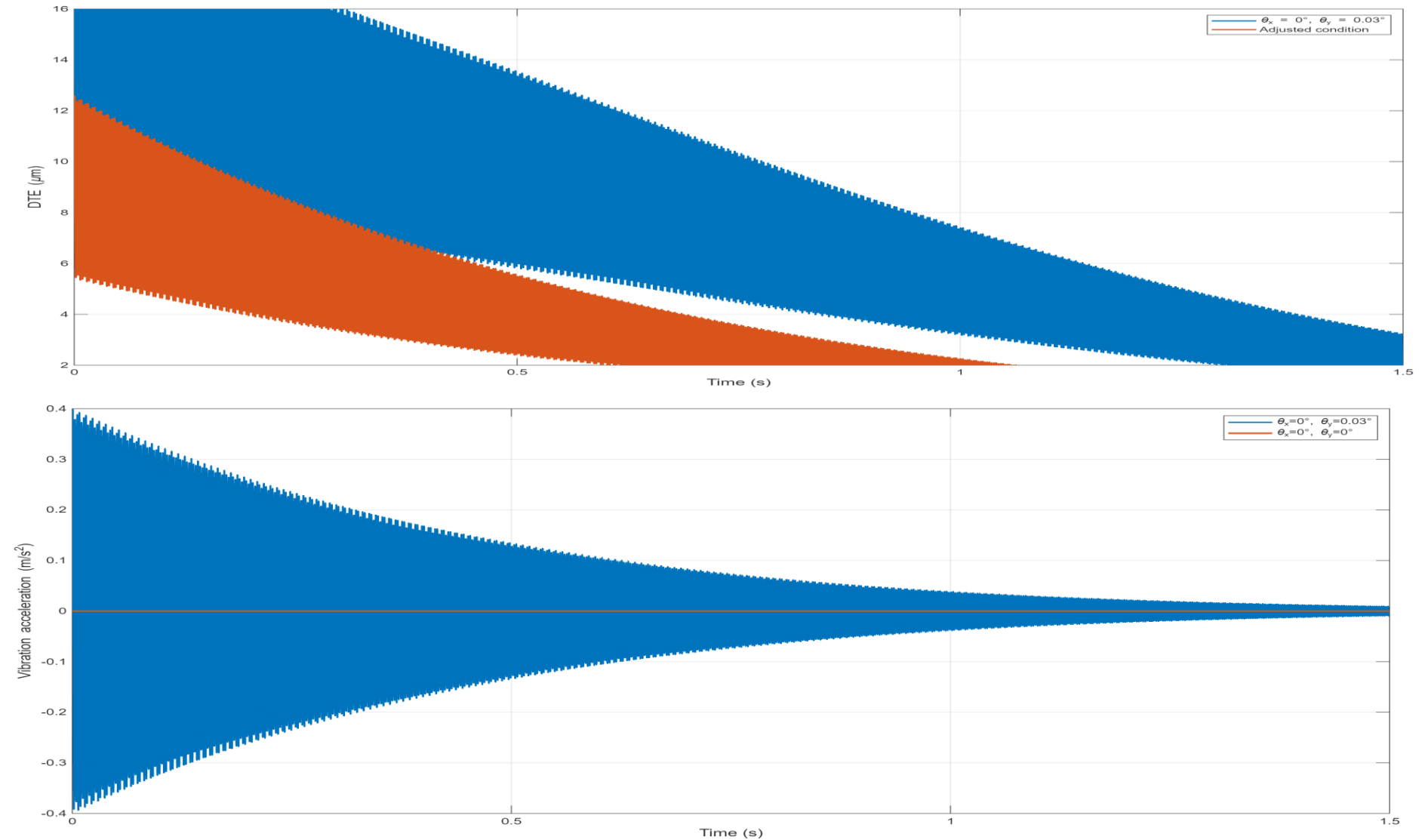


Fig21c , d-Adjustment process of angular misalignment error in y direction

## Actual code

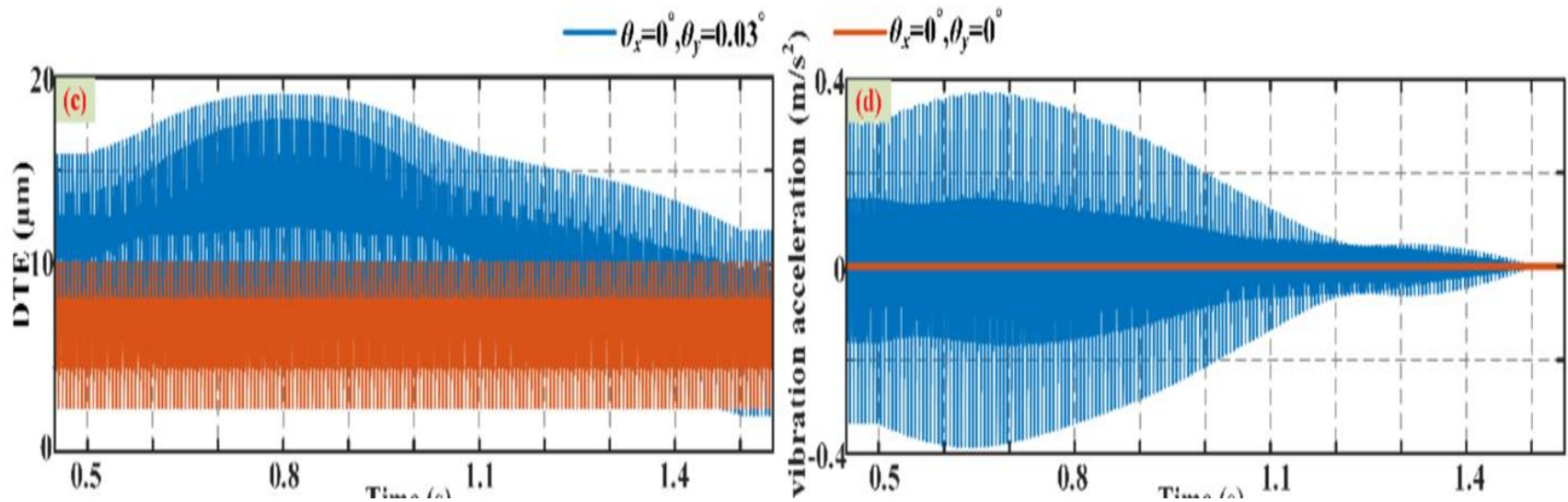
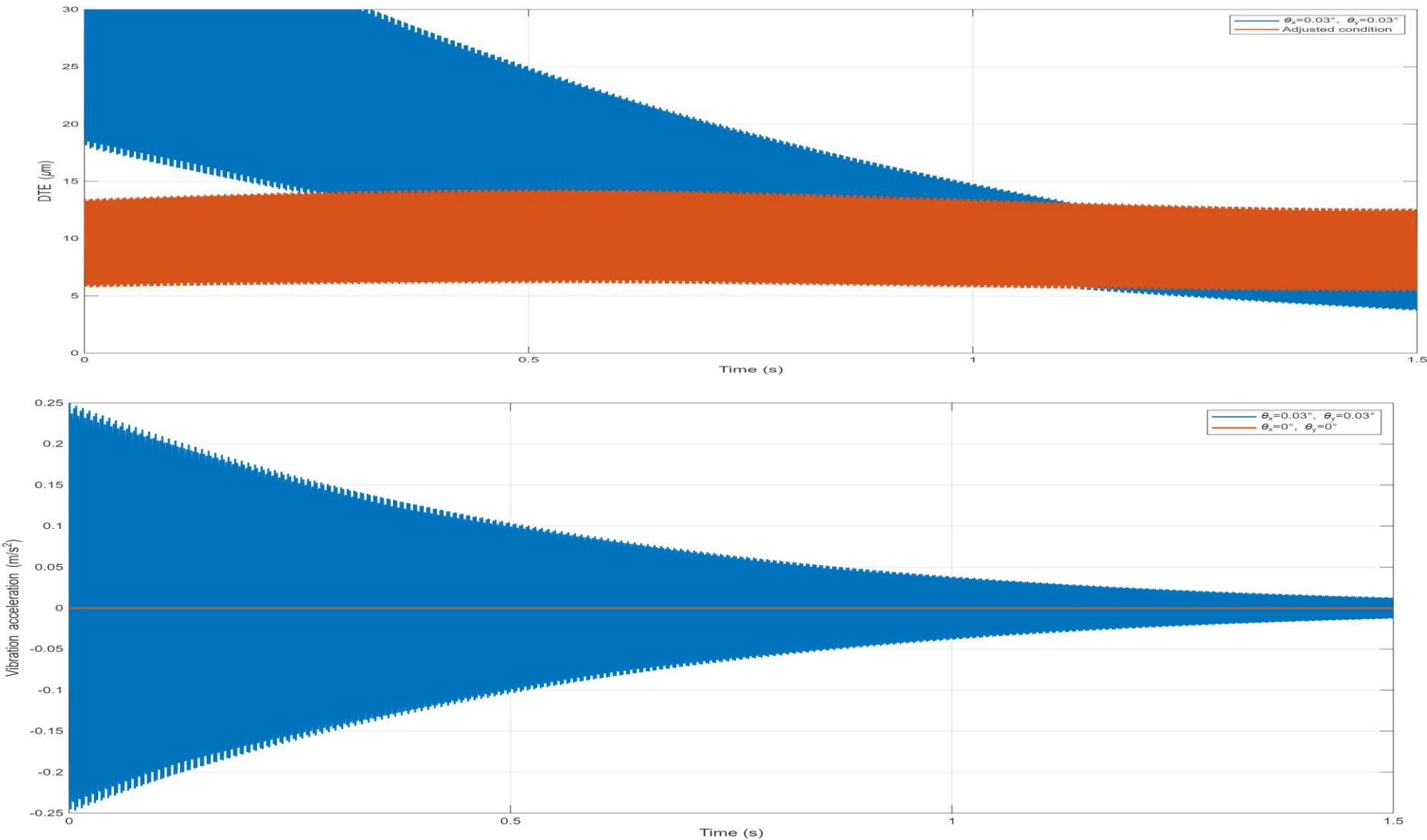


Fig. 21. Adjustment process of angular misalignment error in different direction. (c), (d) y direction



- Fig21e,f-Adjustment process of angular misalignment error in z direction

# Actual Code

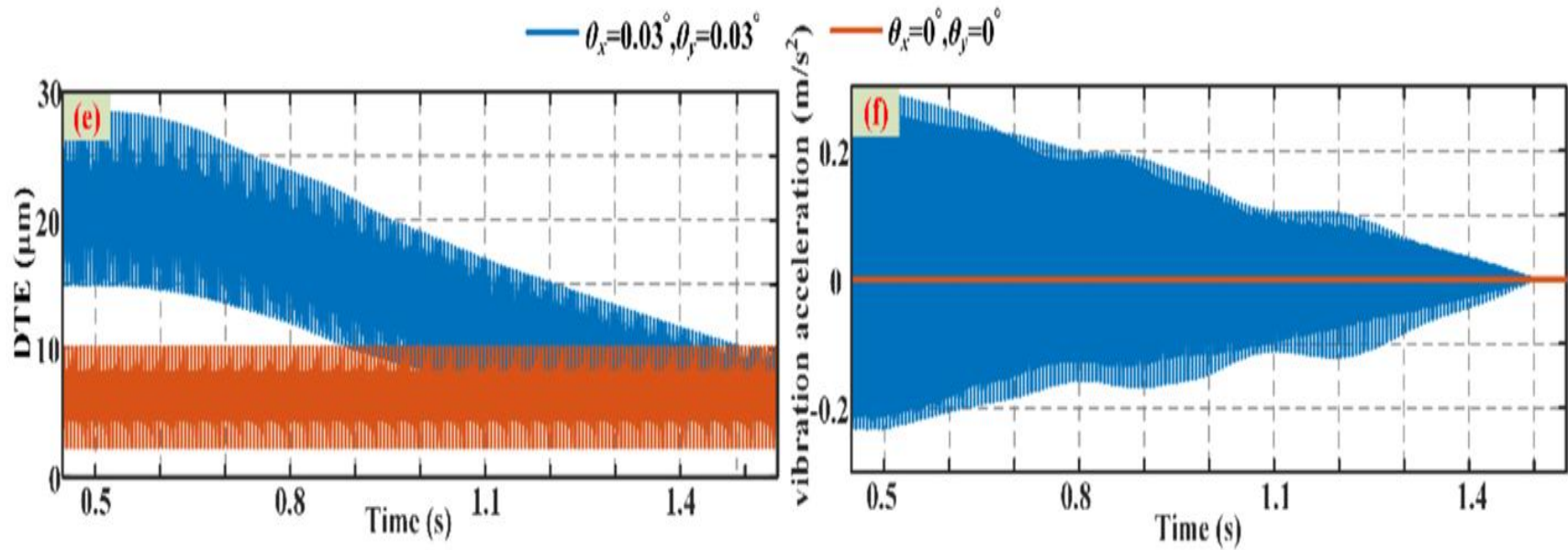
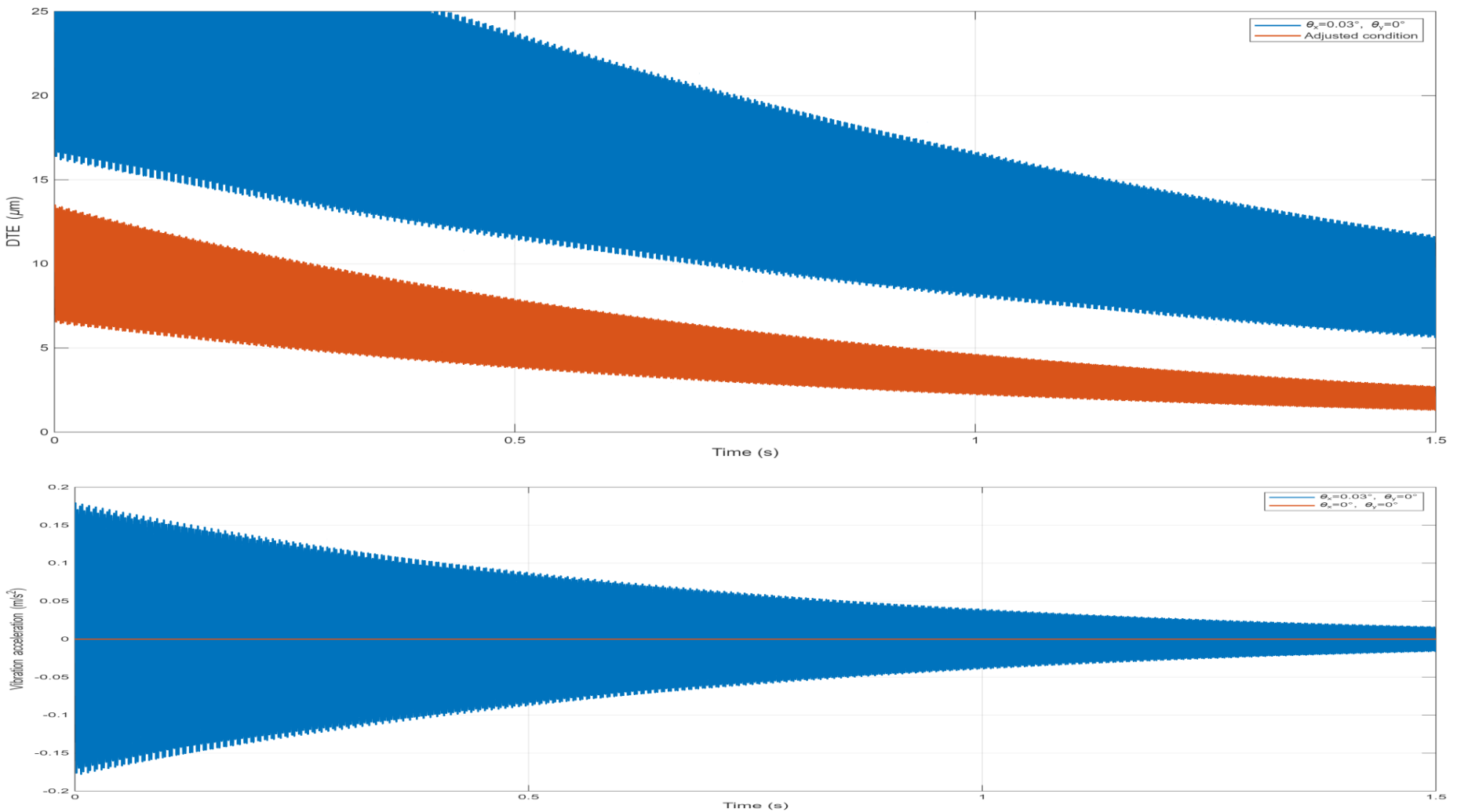


Fig. 21. Adjustment process of angular misalignment error in different direction. (e), (f) comprehensive direction

# Matlab Plot



- Fig22a,b-Adjustment process of angular misalignment error in x direction

## Actual Plot

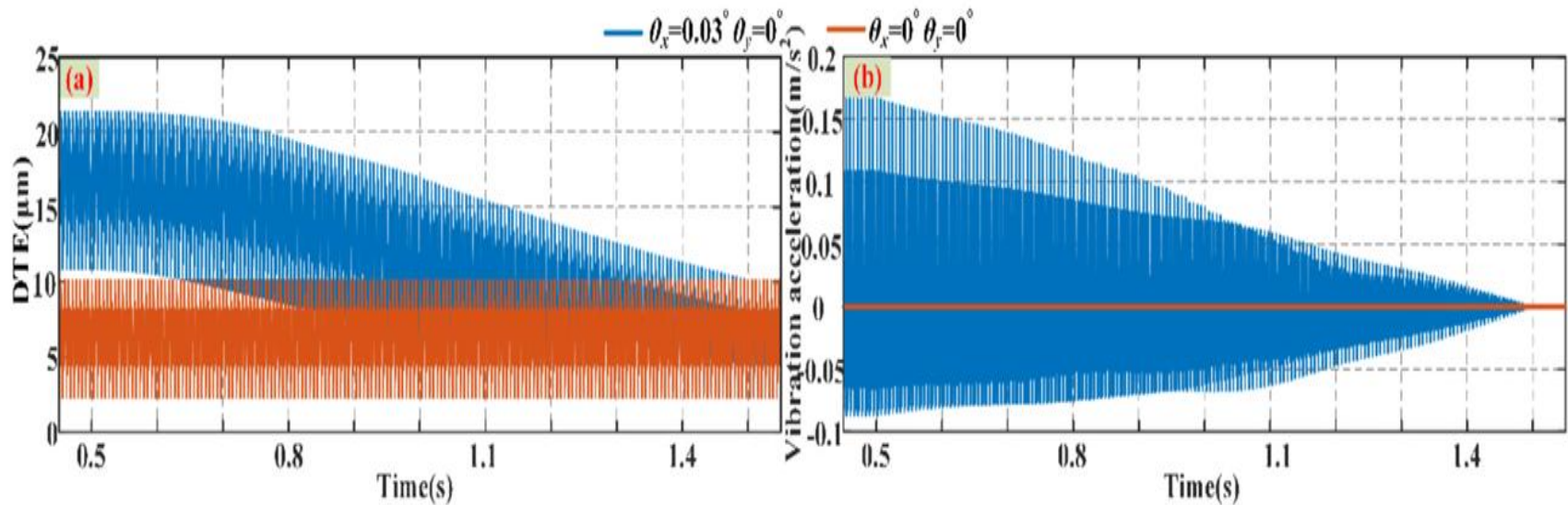
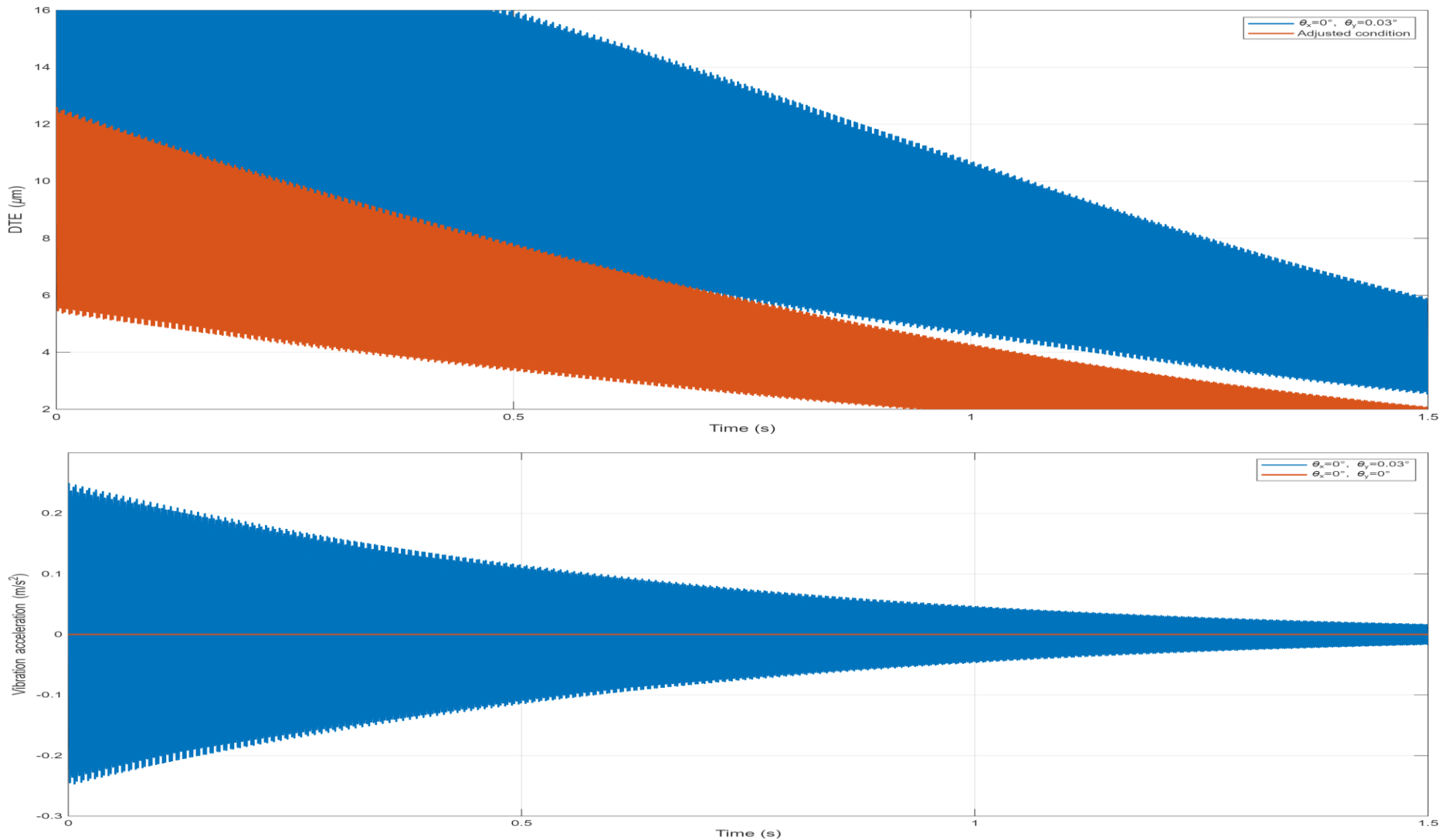


Fig. 22. Adjustment process of angular misalignment error in different direction.  
(a), (b) x direction

# Matlab Code



- Fig22c,d-Adjustment process of angular misalignment error in y direction

## Actual Plot

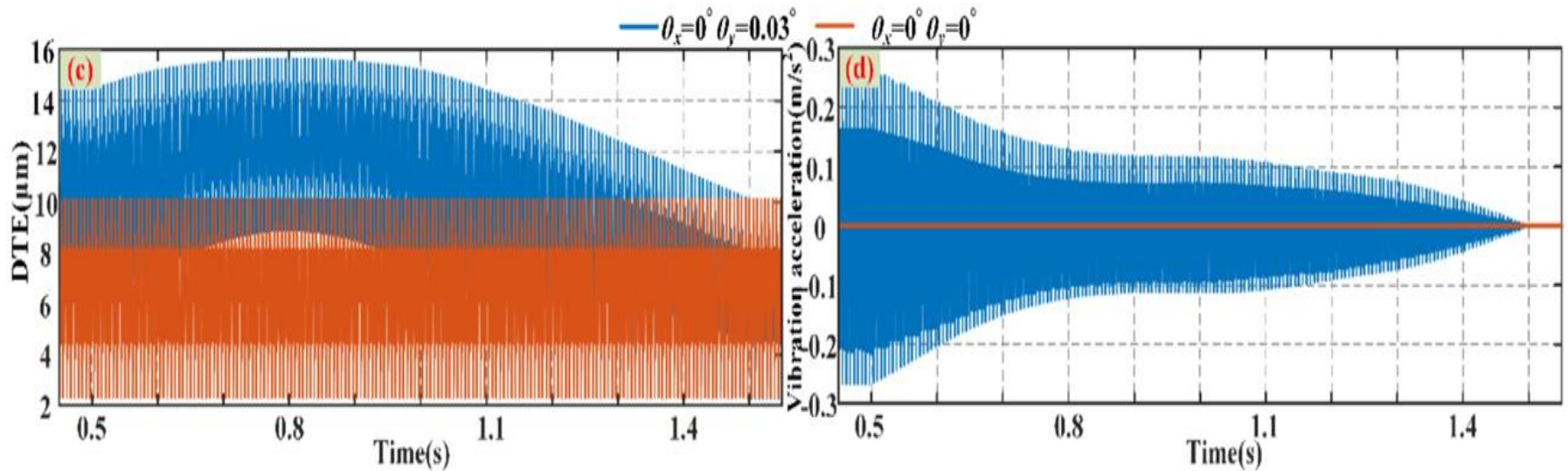
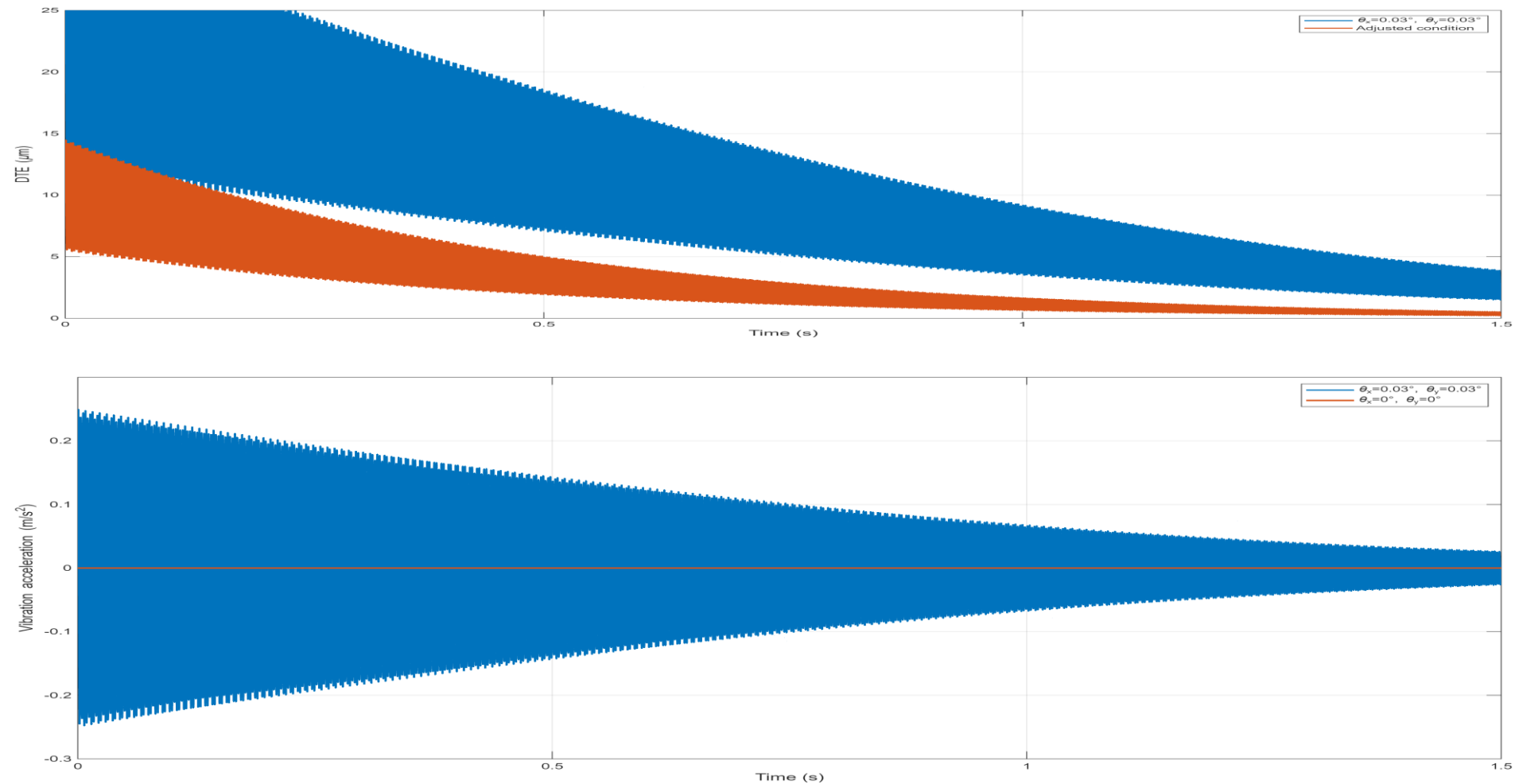


Fig. 22c,d Adjustment process of angular misalignment error in different direction. (c), (d) y direction

## Matlab Plot



- Fig22e,f-Adjustment process of angular misalignment error in z direction

## Actual plot

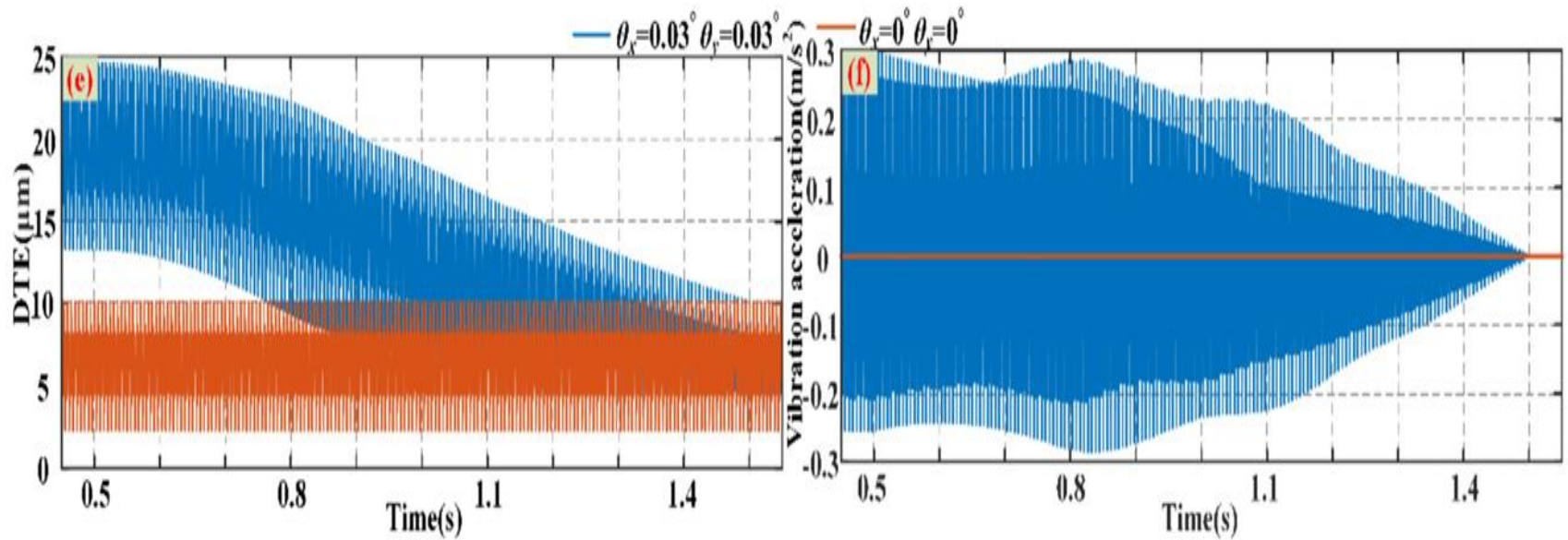


Fig22e,f-Adjustment process of angular misalignment error in z direction

## Matlab Code

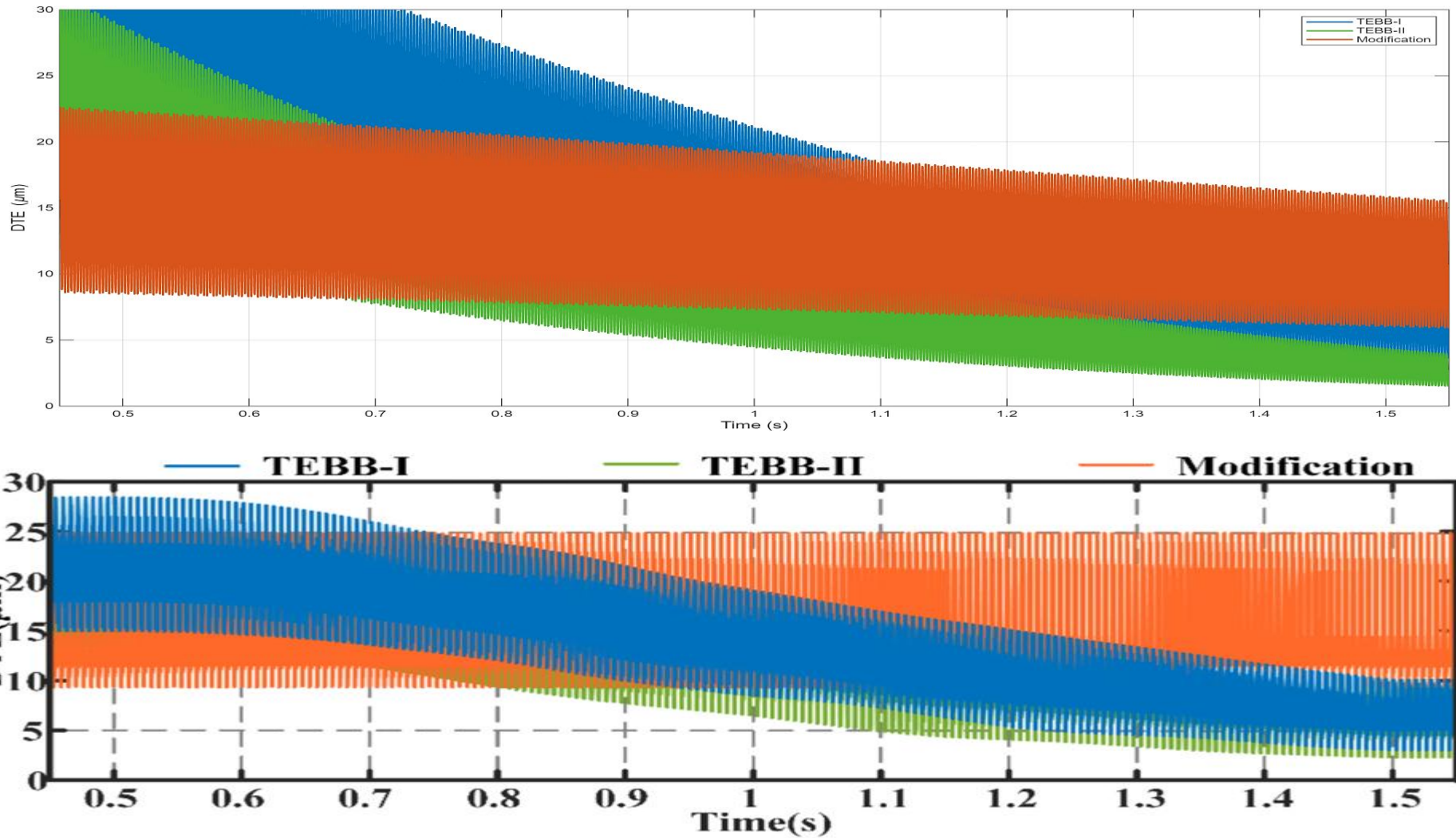


Fig. 23. Comparison between the proposed control method from a comprehensive perspective and crown modification method.

# Discussion

- Strengths of TEBB-I:
  - Simple and cost-effective design
  - Suitable for automotive transmissions and agricultural systems etc.
- Strengths of TEBB-II:
  - Wide adjustment range
  - Ideal for precision machinery and CNC systems(robots)

# References

- Zou, D., Wang, L., Zou, B., Yu, W., Shao, Y., & Mechefske, C. (2025).
- Y.H. Hu, Q.G. Du, S.H. Xie, Nonlinear dynamic modeling and analysis of spur gears considering uncertain interval shaft misalignment with multiple degrees of freedom[J], Mech. Syst. Sig. Process. 193 (2023) 110261.